

# Andres M. Arias Lorza, PhD

Wroclaw, Poland | arias@zib.de | +48-573454806 | xandresariasx.github.io

## Summary

---

Experienced managing biomedical image analysis, machine learning, and data analysis projects.

15 years experience supervising multidisciplinary teams of researchers in the development of biomedical imaging and data analysis research projects. 5 years experience in pharmaceutical research supervising projects related to clinical trials from new experimental therapies.

My overall knowledge of methodologies in imaging and data analysis, together with my technical knowledge in multiple programming languages and tools, and experience supervising research projects makes me an ideal candidate to manage and supervise research projects from its conception, development, validation, and manufacturing.

## Experience

---

**Associate Researcher**, Zuse Institute Berlin – Berlin, DE May 2022 – Present

- Develop a tool for knee image segmentation using U-Nets and transfer learning
- Developed dynamic knee image reconstruction method, enabling tissue deformation analysis during motion
- Tutored PhD and masters students

**Postdoctoral Researcher**, Moffitt Cancer Center – Tampa, FL Jan 2018 – Apr 2022

- Created quantitative imaging tools to evaluate cancer treatment responses in clinical trials
- Identified imaging biomarkers predicting treatment response with >70% accuracy

**Doctoral Researcher**, Erasmus University – Rotterdam, NL Oct 2011 – Oct 2017

- Developed novel segmentation and registration methods for carotid artery imaging with >90% segmentation overlap and sub-millimeter registration accuracy
- Applied image analysis tools in big population studies resulting in discoveries in the mechanics of the arteries and stroke prediction

## Education

---

**Erasmus University**, PhD in Biomedical Image Processing Oct 2011 – Oct 2017

- Thesis: *Image analysis of the carotid artery: a (semi-)automatic approach*

**Eindhoven University of Technology**, Master in Embedded Systems Aug 2009 – Sep 2011

- Thesis: *Analysis of 3D MRI Blood-Flow Data using Helmholtz Decomposition*

**Pontificia Universidad Javeriana**, BS in Electronics Engineering Jan 2001 – Oct 2007

- Thesis: *Voice recognition system implemented in a Digital Signal Processor (DSP)*

## Selected Publications

---

**Magnetic resonance imaging of tumor response to stroma-modifying pegvorhyaluronidase alpha (PEGPH20) therapy in early-phase clinical trials** 2024

Arias et al., Nature Scientific Reports, 10.1038/s41598-024-62470-9

**Dose-response assessment by quantitative MRI in a phase 1 clinical study of the anti-cancer vascular disrupting agent crolibulin** 2020

Arias et al., Nature Scientific Reports, 10.1038/s41598-020-71246-w

**Carotid Artery Wall Segmentation in Multispectral MRI by Coupled Optimal Surface Graph Cuts** 2015

Arias et al., IEEE Transactions on Medical Imaging, 10.1109/TMI.2015.2501751

## Projects

---

**HR segmentation of LR images from public challenge datasets** GitHub

- Develop a tool to obtain high resolution segmentation of structures from low resolution and poor contrast images
- Tools Used: Batch, Python, Jupyter notebooks, AWS

**Diffusion and perfusion parameters from MRI images** GitHub

- Developed a tool that generates ADC maps from DW-MRI data; T1 maps from T1-MRI; and Ktrans; Vp, Ve, iAUC maps from DCE-MRI
- Tools Used: Matlab, multi-core programming

**Tool to analyze the knee movement from dynamic imaging** Technical Report

- Developed a tool to obtain high resolution reconstruction of dynamic low resolution images of the knee making possible to analyze bones and soft tissue deformations during movement
- Tools Used: Python, Matlab, high-performance computing programming

**Tool to segment the carotid arteries from MRI** Technical Report

- Developed a tool to segment the arteries from MRI using a graph-cut approach
- Tools Used: C++, Python, Matlab, high-performance computing programming

## Technologies

---

**Languages:** C++, Python, Matlab, Batch, high-performance computing programming, HTML

**Technologies:** Amira, Mevislab, 3D slicer, Wolfram Mathematica, Amazon Web Services, Jupyter Notebooks, Weka machine learning tool, Elastix image registration tool

## Courses & Certifications

---

**Machine Learning Operations (MLOps)** Coursera Online Course

**Intro to TensorFlow for Deep Learning** Udacity Online Course

**Deep Learning with MATLAB** Mathworks Online Course

**Medical Imaging Summer School** Favignana, IT

**Scientific English Writing** Rotterdam, NL

**Computer Vision Summer School** Ragusa, IT

**Biomedical Image Analysis Summer School** Paris, FR

**Advanced Pattern Recognition** Delft, NL

**Knowledge Driven Image Segmentation** Leiden, NL

**Front end Vision & Multi-Scale Image Analysis** Eindhoven, NL

**ISMRM conference X 2** Virtual Meeting

**ISMRM conference** Montreal, CA

**MICCAI conference** Nagoya, JP

**MICCAI conference** Nice, FR